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# Can Tobacco Control Policies Prevent Tooth Loss? A Difference-in-Differences Study

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- Smoking is a modifiable cause of periodontitis and tooth loss
- Cigarette taxes and prices reduce smoking; the evidence for smoke-free laws is weaker
- Evidence linking tobacco-control policies to oral health outcomes at population scale is limited

### **Primary aim (substantive)**

Estimate the effect of U.S. state tobacco-control policies on smoking and on tooth loss

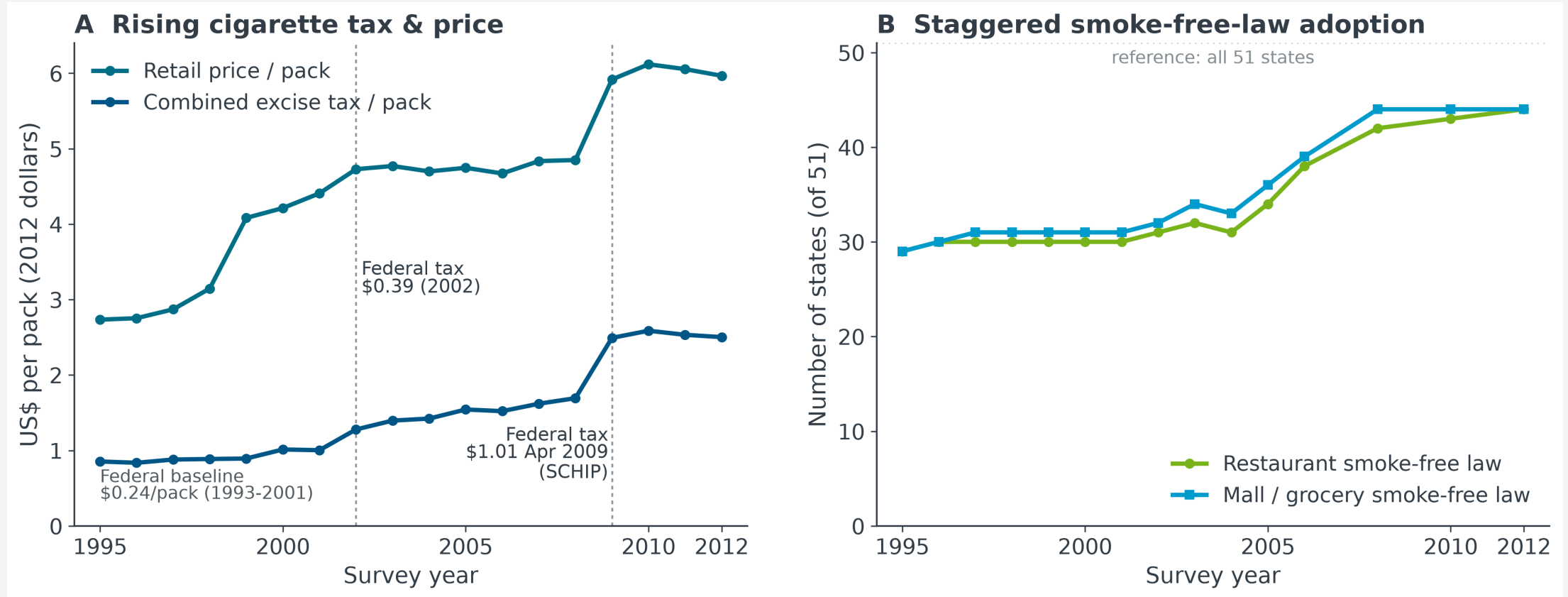
### **Secondary aim (methodological)**

Staggered binary and continuous treatments: compare two-way fixed effects with heterogeneity-robust DiD estimators

|                         | Operationalisation                                                                                                                               |
|-------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Sample</b>           | Behavioral Risk Factor Surveillance System (BRFSS), 1995–2012; 15 cycles, 51 states; analytic N = 2,622,010 (smoking) and 2,638,355 (tooth loss) |
| <b>Treatments</b>       | Cigarette tax and price (\$/pack, continuous); restaurant and mall/grocery smoke-free laws (staggered)                                           |
| <b>Outcomes</b>         | Complete edentulism; loss of $\geq 6$ teeth                                                                                                      |
| <b>Control outcomes</b> | Positive: current smoking, quitting; negative: seat-belt use, smoke-detector                                                                     |

# U.S. state tobacco-control policies, 1995–2012

A rising cigarette tax and price (left) and smoke-free-law adoption (right) — the variation the difference-in-differences design exploits



$$Y_{ist} = \beta D_{st} + X'_{ist}\gamma + \alpha_s + \lambda_t + \varepsilon_{ist}$$

i individual, s state, t year. D = policy (dose in \$/pack, or ban vs none); X = covariates;  $\alpha$  = state fixed effects;  $\lambda$  = year fixed effects.  $\beta \times 100$  is in percentage points — per +\$1.00 of dose, or for ban versus no ban

Target estimand: the average treatment effect on the treated, identified under parallel trends in the mean untreated potential outcome

## Staggered binary treatment — the smoke-free laws

- $\beta$  is a weighted average of every observed 2x2 DiD; the weights are non-negative (Goodman-Bacon 2021), but already-treated states serve as controls — contaminated comparisons
- Rewritten over group-time effects the weights need not be non-negative (de Chaisemartin–D'Haultfœuille 2020): the sign of the estimand is not guaranteed

## Continuous dose — cigarette tax and price

- Dose in constant 2012 dollars: every state's real dose moves every year — no stayers (de Chaisemartin et al. 2024)
- No untreated cells after 1996: year effects for 1997–2012 unidentified, so the secular decline is absorbed into the dose coefficient

$$ATT(d_j | d_j) - ATT(d_{j-1} | d_{j-1}) = \underbrace{ACRT(d_j | d_j) \cdot \Delta d}_{\text{causal}} + \underbrace{ATT(d_{j-1} | d_j) - ATT(d_{j-1} | d_{j-1})}_{\text{selection bias}}$$

$d_j$  = dose level (\$/pack),  $\Delta d = d_j - d_{j-1}$ . ATT = average treatment effect on the treated:  $ATT(d | d')$  is the effect of dose  $d$  among states that actually received  $d'$ . ACRT = average causal response on the treated:  $ACRT(d_j | d')$  is a slope — their response per +\$1.00 of dose at  $d_j$  — so  $ACRT \cdot \Delta d$  is the response to moving from  $d_{j-1}$  to  $d_j$ . Comparing dose levels therefore returns that causal response plus selection bias (Callaway et al. 2025)

| Family                                                                     | Comparison group                                                                | What it needs                                                             | What it gave here                                                                                                  |
|----------------------------------------------------------------------------|---------------------------------------------------------------------------------|---------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------|
| <b>Group-time ATT</b><br><b>Callaway–Sant'Anna 2021; Sun–Abraham 2021</b>  | Not-yet-treated (CS); the last-treated cohort where no never-treated exist (SA) | No anticipation + parallel trends; the two rest on non-nested assumptions | Positive and significant on the high-tax cohort — but fails its own positive control                               |
| <b>Imputation</b><br><b>Borusyak–Jaravel–Spiess 2024; Gardner 2022</b>     | All untreated cells, which fit the $Y(0)$ model                                 | Untreated cells in every period, or the year effects are unidentified     | Null throughout; the most stable family on this panel                                                              |
| <b>Extended TWFE / two-way</b><br><b>Mundlak</b><br><b>Wooldridge 2025</b> | Never-treated — saturated, it is a set of 2×2 DiDs against the never-treated    | Cohort × period interactions; heterogeneous effects allowed, not required | Null — but fitted linear in dose, so a specification check; saturated it is algebraically the imputation estimator |

$$ATT(g, t) = E[Y_t(g) - Y_t(0) \mid G_g = 1]$$

The group-time estimand — *identified*, not defined, by the difference-in-differences contrast against a clean comparison cohort, under no anticipation and parallel trends. Continuous dose handled as high-dose cohorts (tax ≥ \$3, price ≥ \$6 per pack)

No clean comparison for the continuous dose: the baseline-deviation specification implies −3.1 pp per \$1.00 on smoking — roughly seven times the adequately identified estimate

Smoking — percentage points (95% CI) per +\$1.00/pack

| Outcome         | Tax, current year    | Tax, 10-year average |
|-----------------|----------------------|----------------------|
| Current smoking | −0.35 (−0.56, −0.13) | −0.47 (−0.95, 0.01)  |
| Quitting        | 0.51 (0.19, 0.82)    | 0.07 (−0.53, 0.67)   |

Tooth loss — percentage points (95% CI) per +\$1.00/pack

| Outcome             | Tax, current year   | Tax, 10-year average |
|---------------------|---------------------|----------------------|
| Complete edentulism | 0.08 (−0.14, 0.31)  | 0.39 (−0.12, 0.89)   |
| Loss of ≥6 teeth    | −0.19 (−0.53, 0.15) | −0.44 (−1.17, 0.30)  |

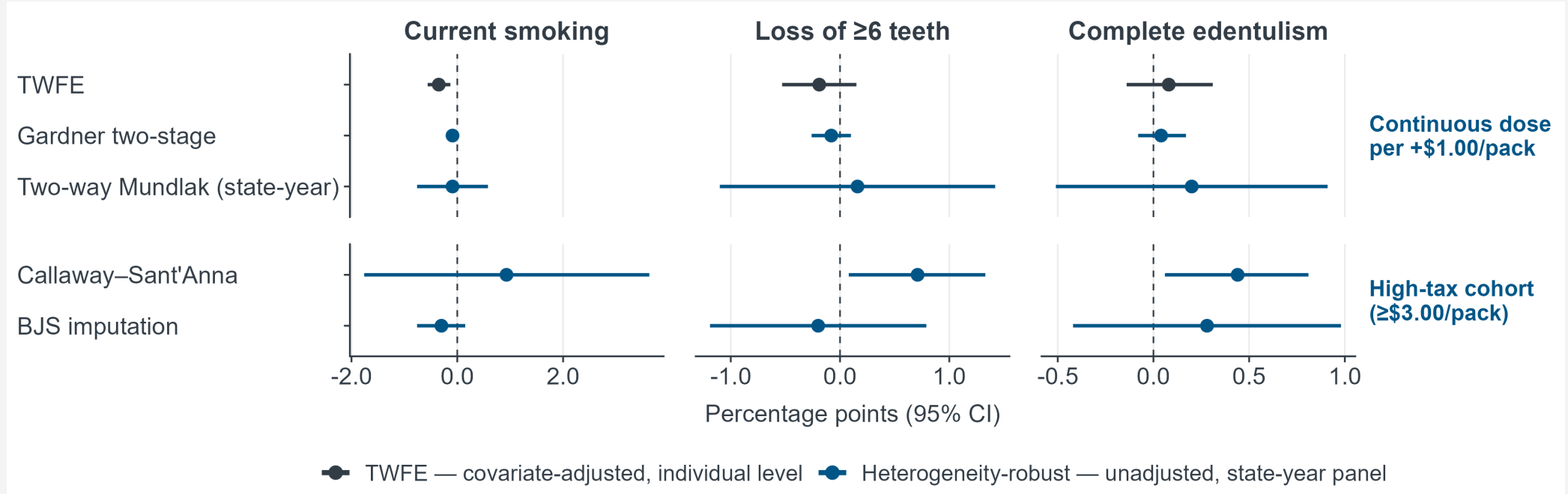
Covariate-adjusted two-way (state and year) fixed-effects estimates; standard errors clustered by state. Retail price shows similar estimates.

Percentage points (95% CI), ban vs no ban

| Ban            | Current smoking    | Complete edentulism | Loss of $\geq 6$ teeth |
|----------------|--------------------|---------------------|------------------------|
| Restaurant     | 0.11 (−0.34, 0.55) | 0.45 (0.03, 0.88)   | 0.51 (−0.02, 1.05)     |
| Mall / grocery | 0.30 (−0.21, 0.81) | 0.24 (−0.19, 0.68)  | 0.52 (−0.15, 1.18)     |

- No first stage: neither ban lowered smoking, and several tooth-loss estimates are positive
- The mall/grocery ban fails its negative control: smoke-detector test −1.17 pp (−2.27, −0.08)
- The tax passes both negative controls (both null)

## Estimator comparison

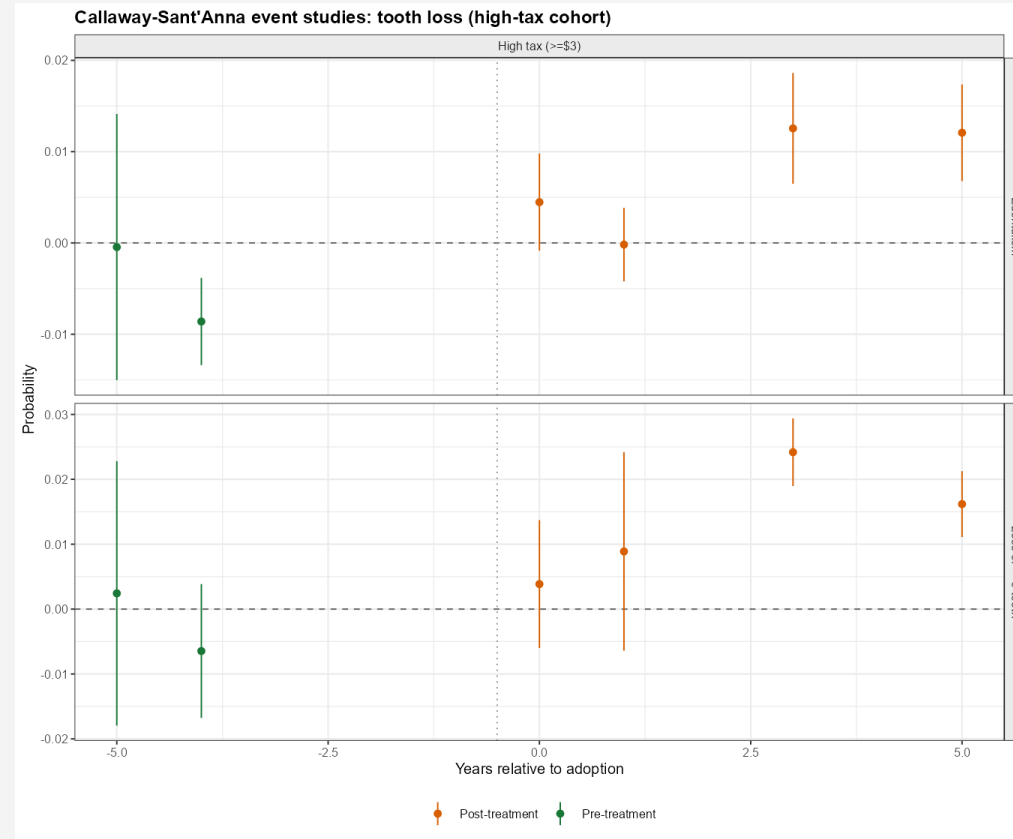


- Left panel: no robust estimator recovers the first stage on the state-year panel — but only Callaway–Sant'Anna gets the sign wrong (+0.93 pp, -1.76 to 3.63), and only it turns positive for tooth loss
- Different estimands: continuous per +\$1.00 dose versus binary  $\geq \$3.00$ -cohort ATT
- Callaway–Sant'Anna rests on small, late, biennially observed cohorts; imputation, more stable, stays null

Sun–Abraham omitted here: for the high-dose tooth-loss cohorts — small, late and biennially observed — its variance is undefined (the group-time estimator does not converge there either); it is estimable for the bans

# Callaway–Sant'Anna event study (Roth 2026)

Universal base period (Roth 2026). Pre-treatment coefficients are not negligible —  $-0.6$  to  $-0.9$  pp, larger than any effect of interest — and post-treatment coefficients turn significantly positive. Read together: a differential trend, not a protective policy



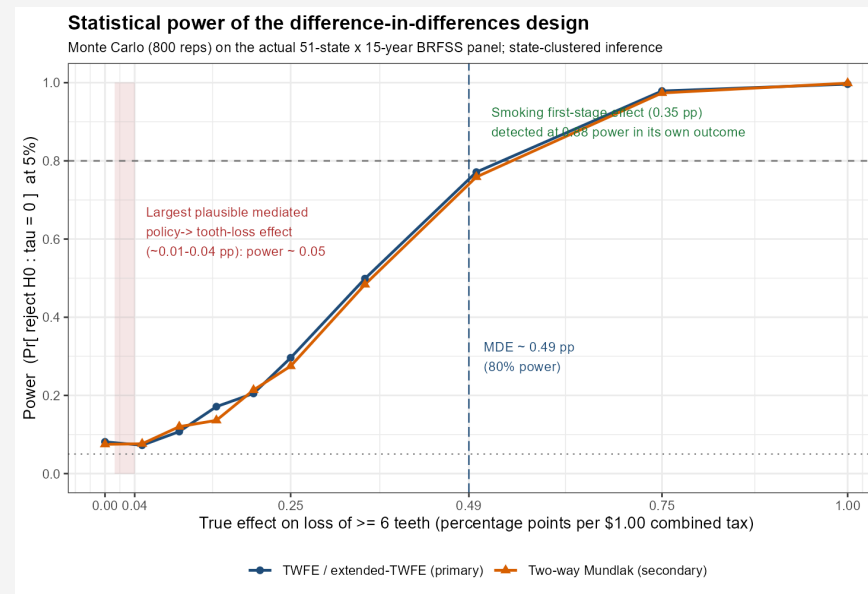
No  $t-1$  (universal base period); no  $t-3$  or  $t-2$  (the oral-health module ran annually to 2006, then every second year); and the high-dose cohorts adopt late

$$E[\Delta Y(0) \mid D = 1] = E[\Delta Y(0) \mid D = 0]$$

- A pre-trend of the size just shown is exactly what this assumption forbids
- Untestable: pre-trend tests have low power, and conditioning on passing one distorts inference (Roth 2022) — judge magnitude
- Identification also needs a stable-composition condition; both are conditions on population means, not individual trajectories
- Rambachan–Roth (2023) relative magnitudes ( $\bar{M} = 2$ ): the high-tax event-study interval for  $\geq 6$  teeth widens from  $[-0.20, 1.50]$  to  $[-9.0, 11.1]$  pp

# Is the null informative? Minimum detectable effect and plausibility

- CI excludes protective effects beyond  $\approx 0.5$  pp for the current-year tax,  $\approx 1.2$  pp for the 10-year average — a precise null, and an equivalence statement rather than evidence of absence (Rainey 2014; Lakens et al. 2018)
- Minimum detectable effect 0.49 pp, against a plausible mediated effect of 0.01–0.04 pp — 10–40 $\times$  below it
- The ceiling is between-state variation and 51 clusters, not the 2.6 m respondents (Bertrand et al. 2004)
- Detecting even the most generous plausible effect (0.04 pp) would take  $\approx 7,700$  states, not 51 — more respondents or more years do not close that gap (backup)



1. Cigarette taxes and prices reduce smoking; we find no detectable effect on tooth loss
2. That null is a limit of the design, not a finding about teeth
3. The robust estimators diverge — but that is a small-sample and parallel-trends result, not a substantive one: the estimator that turns positive fails both its positive control and its pre-trends
4. Continuous-dose difference-in-differences is the binding constraint here: no stayers, no untreated cells, and dose contrasts that carry selection bias



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# Thank you

Paper in print in the *Journal of Dental Research*

Slides and code:

[github.com/BaumeisterS/DGEPI2026\\_tobacco\\_policy\\_tooth\\_loss](https://github.com/BaumeisterS/DGEPI2026_tobacco_policy_tooth_loss)

**DiD estimators:** Goodman-Bacon A. 2021. Difference-in-differences with variation in treatment timing. *J Econom.* 225(2):254–277. de Chaisemartin C, D'Haultfœuille X. 2020. Two-Way Fixed Effects Estimators with Heterogeneous Treatment Effects. *Am Econ Rev.* 110(9):2964–2996. Callaway B, Sant'Anna PHC. 2021. Difference-in-Differences with multiple time periods. *J Econom.* 225(2):200–230. Sun L, Abraham S. 2021. Estimating dynamic treatment effects in event studies with heterogeneous treatment effects. *J Econom.* 225(2):175–199. Borusyak K, Jaravel X, Spiess J. 2024. Revisiting Event-Study Designs: Robust and Efficient Estimation. *Rev Econ Stud.* 91(6):3253–3285. Gardner J. 2022. Two-stage differences in differences. *arXiv:2207.05943*. Wooldridge JM. 2025. Two-way fixed effects, the two-way Mundlak regression, and difference-in-differences estimators. *Empir Econ.* 69(5):2545–2587. de Chaisemartin C, D'Haultfœuille X, Vazquez-Bare G. 2024. Difference-in-Difference Estimators with Continuous Treatments and No Stayers. *AEA Pap Proc.* 114:610–613. Callaway B, Goodman-Bacon A, Sant'Anna PHC. 2025. Difference-in-differences with a Continuous Treatment. *arXiv:2107.02637v8* (also NBER Working Paper 32117).

**Identification, sensitivity and reviews:** Roth J. 2022. Pretest with Caution: Event-Study Estimates after Testing for Parallel Trends. *Am Econ Rev Insights.* 4(3):305–322. Roth J. 2026. Interpreting event-studies from recent difference-in-differences methods. *Jpn Econ Rev.* 77(2):275–288. Rambachan A, Roth J. 2023. A More Credible Approach to Parallel Trends. *Rev Econ Stud.* 90(5):2555–2591.

**Power, precision and equivalence:** Bertrand M, Duflo E, Mullainathan S. 2004. How Much Should We Trust Differences-In-Differences Estimates? *Q J Econ.* 119(1):249–275. Rainey C. 2014. Arguing for a Negligible Effect. *Am J Polit Sci.* 58(4):1083–1091. Lakens D, Scheel AM, Isager PM. 2018. Equivalence Testing for Psychological Research: A Tutorial. *Adv Methods Pract Psychol Sci.* 1(2):259–269.

## Backup: the four 2×2 comparisons under staggered adoption

TWFE averages over every 2×2 difference-in-differences the data allow. Three are clean; the fourth is not

|   | Treated group  | Control group                                           | Clean? |
|---|----------------|---------------------------------------------------------|--------|
| 1 | Early adopters | Never-treated                                           | Yes    |
| 2 | Late adopters  | Never-treated                                           | Yes    |
| 3 | Late adopters  | Early adopters, before they adopt                       | Yes    |
| 4 | Early adopters | Late adopters, while the early group is already treated | No     |

In comparison 4 the control group is already treated, so its outcome change carries its own evolving treatment effect rather than a clean counterfactual trend. Subtracting it subtracts part of an effect — which is why the bias term is non-zero only when treatment effects change over time

## Backup: what TWFE estimates under staggered adoption

The two decompositions behind slide 5 — the weights are non-negative in the first and need not be in the second

$$\hat{\beta}^{\text{TWFE}} = \sum_k s_k \widehat{\text{DD}}_k, \quad s_k \geq 0, \quad \sum_k s_k = 1$$

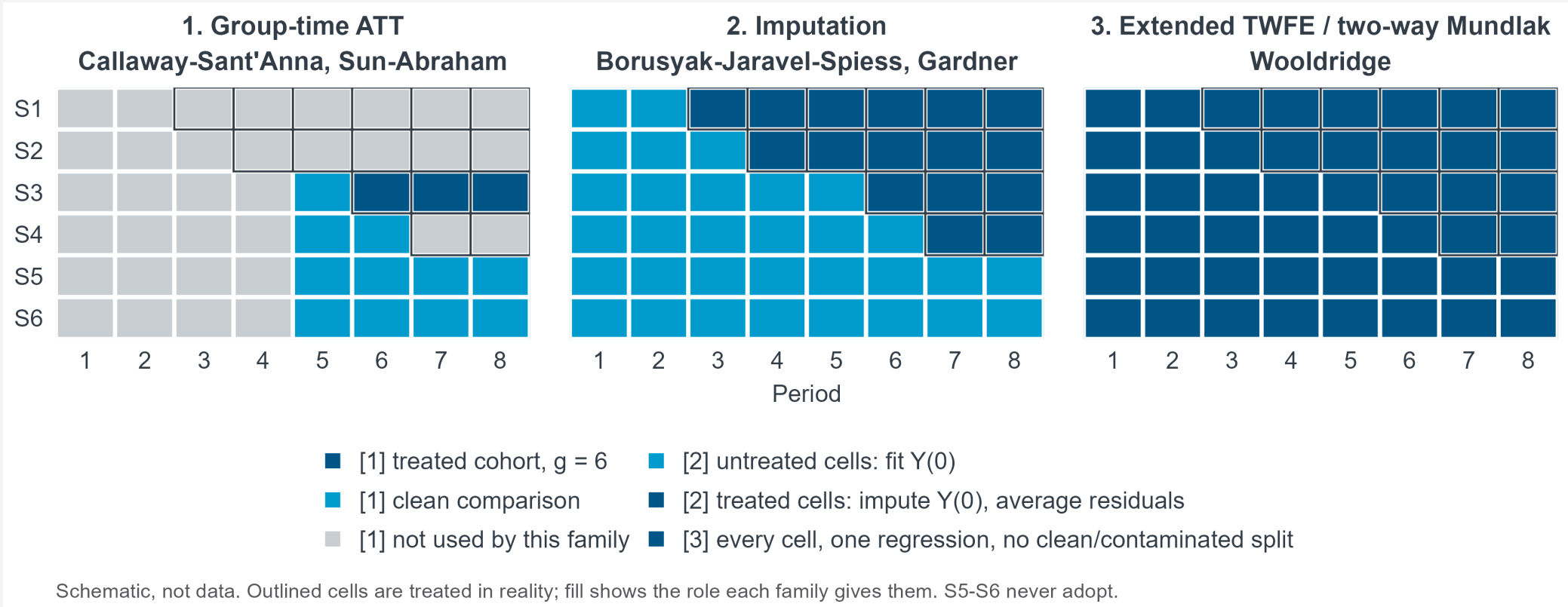
Goodman-Bacon (2021): a variance-weighted average of all observed 2×2 DiDs. The weights are variance shares, so they cannot be negative; the bias enters because some 2×2 comparisons use already-treated units as controls

$$\beta^{\text{TWFE}} = \sum_{g,t} w_{g,t} \text{ATT}(g,t), \quad \exists w_{g,t} < 0$$

de Chaisemartin–D'Haultfœuille (2020): rewritten over group-time effects, the weights need not be non-negative — so  $\beta$  can carry the wrong sign even when every underlying  $\text{ATT}(g,t)$  has the same sign

# Backup: which panel cells each family uses

The same three families, drawn on a stylised panel with staggered adoption



## Backup: inference with 51 clusters

- Standard errors clustered by state throughout (51 clusters)
- Monte Carlo on the actual 51-state  $\times$  15-year panel (800 replications): power  $\approx 0.50$ , 0.77 and 0.98 at true effects of 0.35, 0.50 and 0.75 pp
- The simulation shows a mild finite-cluster over-rejection (size  $\approx 0.08$ ) — conservative for a no-effect conclusion
- A CR2 bias-reduced, Satterthwaite-corrected small-cluster variance leaves the intervals essentially unchanged, confirming the 51-cluster inference is not anti-conservative
- The equivalence bound is read from the interval itself, not from the simulated size

## Backup: what design would detect the plausible effect?

Suppl. Table 1 gives  $SE = 0.174$  pp and  $MDE = 0.49$  pp for loss of  $\geq 6$  teeth, i.e.  $MDE = 2.8 \times SE$  (80% power, 5% two-sided). Since the ceiling is between-state variation and the number of clusters, precision scales as  $1/\sqrt{G}$  — so the states required are  $51 \cdot (0.49/\text{target})^2$

| Target effect                              | SE needed | States needed     | vs the 51 available |
|--------------------------------------------|-----------|-------------------|---------------------|
| <b>0.04 pp — most generous plausible</b>   | 0.014 pp  | $\approx 7,700$   | 150×                |
| <b>0.013 pp — central chained estimate</b> | 0.005 pp  | $\approx 72,000$  | 1,400×              |
| <b>0.01 pp — lower end</b>                 | 0.004 pp  | $\approx 122,000$ | 2,400×              |

- More respondents do not help: the binding constraint is between-state variation, not sampling error. Were it otherwise, 0.04 pp would still need  $\approx 400$  million respondents
- More follow-up years buy little precision either — serially correlated state shocks are exactly what Bertrand et al. (2004) show defeats naive time-series gains; longer follow-up raises the effect to be found rather than lowering the threshold

Not formally addressed in the paper — the argument, not a published result

- Interference is plausible: residents of a high-tax state can buy across a state line, so a neighbour's consumption depends on this state's tax
- Direction: cross-border purchasing weakens the realised dose in treated states, attenuating the first stage toward the null
- The first stage is nonetheless detected ( $-0.35$  pp per \$1.00), so attenuation does not overturn it
- For tooth loss the design is already far short of the minimum detectable effect, so interference is not what produces the null